

3-tube AR-forcing analysis v0.1 — is ‘3 generations = 3 exceptional tubes’ FORCED by the Auslander-Reiten structure of $\hat{B}_2$, or counted-to-match?

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3-tube AR-forcing analysis

0. The gate (Keeper #407, “the one I’d watch hardest”)

The bet’s sharpest test: 3 generations = 3 exceptional tubes of the affine quiver $\hat{B}_2$. Cal’s sharpened question: is “3” FORCED by the Auslander-Reiten structure (no freedom in what to count), or counted-to-match? If forced, generation-counting finally turns from matched to derived — the deepest open gate. This doc settles the structural half rigorously and routes the one remaining number to the primary source (per yesterday’s sourcing discipline — I will not guess the tubular type from memory).

1. The structural result (RIGOROUS — settles “forced vs counted”)

For a connected tame hereditary algebra H (path algebra of a Euclidean/affine quiver, or the species version for non-simply-laced types), the module category has the standard trichotomy (Dlab-Ringel, Ringel “Tame algebras and integral quadratic forms”):

$\text{mod } H = (\text{preprojective}) \vee (\text{regular}) \vee (\text{preinjective})$

The regular modules form a tubular family parameterized by $P^1(k)$: - ALMOST ALL tubes are homogeneous (rank 1). - FINITELY MANY (at most 3) are exceptional (rank > 1). - The ranks $(r_1, \dots, r_s)$ of the exceptional tubes — the tubular type — are an INTRINSIC INVARIANT of the algebra, determined by its Auslander-Reiten structure.

This settles the “forced vs counted” question in the bet’s favor — structurally: the exceptional-tube count is NOT a free parameter we tune to hit 3. It is FORCED by the AR-quiver of $\hat{B}_2$ — a fixed representation-theoretic fact about the algebra. Whatever the number is, it is derived, not counted-to-match.

So the question reduces cleanly to: what is the tubular type of $\hat{B}_2$?

2. The one remaining number (ROUTED to the source — NOT guessed)

The exceptional-tube count for $\hat{B}_2$ specifically requires the Dlab-Ringel classification of tame hereditary algebras, INCLUDING the non-simply-laced species cases. I am NOT asserting it from memory (yesterday the genus name flipped three times from memory-relabeling; the fix is to pin to the book).

What I CAN state, with the simply-laced cases as context:

Euclidean type	tubular type	# exceptional tubes
$\tilde{A}_{\{p,q\}}$	(p, q)	2
$\tilde{D}_n$	$(2, 2, n-2)$	3

Euclidean type	tubular type	# exceptional tubes
$\tilde{E}_6$	(2,3,3)	3
$\tilde{E}_7$	(2,3,4)	3
$\tilde{E}_8$	(2,3,5)	3

Observation (suggestive, NOT a proof for $\hat{B}_2$): every non- $\tilde{A}$ *simply-laced* Euclidean type has exactly 3 exceptional tubes. This is what Elie’s Toy 3598 leaned on. BUT $\hat{B}_2$ is non-simply-laced (it is realized as a species / valued quiver, not an ordinary quiver), and the non-simply-laced affine types ($\tilde{B}$, $\tilde{C}$, $\tilde{F}$, $\tilde{G}$) have their own tubular types in the Dlab-Ringel classification that must be checked, not assumed equal to the simply-laced pattern. Some non-simply-laced cases differ.

Routed (Cal + Keeper literature pin): the exact tubular type of $\hat{B}_2$ from Dlab-Ringel “Indecomposable representations of graphs and algebras” (Mem. AMS 1976) or Ringel’s tame-algebras book. This is the single fact that decides the gate.

3. The two outcomes (and why both are valuable)

- If $\hat{B}_2$ ’s tubular type has exactly 3 exceptional tubes: the generation count is FORCED — matched $\rightarrow$ DERIVED. This is the mechanism for the deepest open gate (why 3 generations), and it joins the cyclotomic $\leftrightarrow$ Coxeter / Catalan-Mersenne over-determination cluster as the *representation-theoretic* leg — the cleanest one, because AR structure leaves no freedom.
- If $\hat{B}_2$ ’s tubular type is NOT 3 (e.g., 2, like $\tilde{A}$, or some other count): the “tubes = generations” identification breaks — and that is a clean, valuable break (Keeper’s discipline: a clean break tells us the bet is wrong at a precise point, which is information, not failure). It would mean generations are NOT the tube count and we’d look elsewhere (the cyclotomic chain length, which is independently $4 \rightarrow 3$).

Either way the gate is decisive — which is exactly why it’s worth doing now.

4. Connection to the generation-forcing I’ve been circling

I’ve approached generation-forcing twice before — the cyclotomic $\leftrightarrow$ Coxeter mechanism (chain length = $h(B_2) = 4 \rightarrow 3$) and the rank=2 specialness. Both were “matched, over-determined” but not forced (the cyclotomic $\leftrightarrow$ Coxeter *mechanism* was the multi-week gate). The AR-tube route is a different and potentially stronger lever: it doesn’t go through the cyclotomic chain at all — it reads the generation count directly off the affine algebra’s regular-module structure. If the tube count is 3 and forced, that is the forcing the cyclotomic route was missing — generations come from the AR structure of $\hat{B}_2$, not from a chain-length coincidence.

This also sharpens the bet’s internal consistency check: the cyclotomic chain gives $4 \rightarrow 3$ (chain minus base/seed); the tube count would give 3 directly. If both hold, they must be the *same* 3 — the base/seed ($X=\text{rank}$) in the chain picture should correspond to the homogeneous-tube generic point (not an exceptional tube) in the AR picture. That correspondence is a concrete cross-check (next-step, finite).

5. What I deliver now vs what’s routed

Delivered (rigorous): the structural result — the exceptional-tube count of a tame hereditary algebra is an AR-FORCED intrinsic invariant, not a tunable count. This converts the bet’s “forced vs counted” question: the tube structure is forced; only the specific number is open.

Routed (Cal + Keeper, primary-source pin): $\hat{B}_2$ ’s exact tubular type from Dlab-Ringel — the one fact that decides 3-vs-not.

Next (finite, my lane): if 3 confirmed, the base/seed $\leftrightarrow$ homogeneous-generic correspondence cross-check (cyclotomic-4 vs tube-3 consistency); and identifying which tube = which generation (feeds the particle $\leftrightarrow$ module dictionary + the Periodic Table derived-flags).

6. Honest scope + tier

RIGOROUS (standard rep theory, Dlab-Ringel/Ringel): tame hereditary trichotomy; tubular family over P^1 ; ≤ 3 exceptional tubes; tubular type = intrinsic AR invariant (FORCED). Simply-laced tubular types (table §2).

ROUTED (not asserted): $\hat{B}_2$'s exact tubular type — Dlab-Ringel species classification. Deliberately NOT guessed.

FRAMEWORK (the bet): tubes = generations — the physical identification (which tube $\leftrightarrow$ which generation) needs the particle $\leftrightarrow$ module dictionary (Phase 2). The COUNT being forced is settled structurally; the IDENTIFICATION is earned downstream.

Cal #27 / honesty: the suggestive simply-laced “always 3” pattern is exactly the kind of thing that could mislead — $\hat{B}_2$ is non-simply-laced and may differ. I state the pattern as suggestive context, not proof, and route the decisive number to the book. The structural forcing result is the rigorous deliverable; the number is the gate.

Decisiveness note (Keeper): this is pure representation theory, decisive on its own — it does NOT need the canonical basis or the particle dictionary to settle whether the tube count is forced (it is) and what it is (Dlab-Ringel). It is the gate that could flip the generation count to DERIVED.

— Lyra, 3-tube AR-forcing analysis v0.1 (#407). STRUCTURAL RESULT (rigorous): the exceptional-tube count of $\hat{B}_2$ is an Auslander-Reiten-FORCED intrinsic invariant — NOT counted-to-match; the bet's “forced vs counted” question is settled in favor of forced. The decisive number (is it 3?) is ROUTED to the Dlab-Ringel tame-hereditary-species pin — NOT guessed (non-simply-laced $\hat{B}_2$ may differ from the simply-laced “always 3”). If 3: generation count DERIVED (deepest gate closed). If not: a clean, valuable break. Either way decisive, and it's the representation-theoretic leg the cyclotomic $\leftrightarrow$ Coxeter route was missing.